

- 7 (a)** Newton's law of gravitation states that the force of attraction between two point masses is directly proportional to the product of their masses and inversely proportional to the square of their distance apart.

(b) (i)

$$\begin{aligned} g &= \frac{GM_E}{r^2} = \frac{GM_E}{(R_E + 350 \times 10^3)^2} \\ &= \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{(6.4 \times 10^6 + 350 \times 10^3)^2} \\ &= 8.78 \text{ Nkg}^{-1} \end{aligned}$$

It is along the line joining point P and the centre of the Earth and is pointing towards the Earth's centre.

(ii)

$$F_G = m\omega^2 r \text{ [CONCEPT]}$$

$$gm = m \left(\frac{2\pi}{T} \right)^2 r$$

$$T = \sqrt{\frac{4\pi^2 r}{g}} = \sqrt{\frac{4\pi^2 (6.4 \times 10^6 + 350 \times 10^3)}{8.78}} = 5510 \text{ s}$$

(c) (i) Gravitational potential at a point is defined as work done per unit mass by an external agent to move the unit mass from infinity to that point.

(ii) Since gravitational force is attractive, the force that the external agent exerts (which points towards infinity) is opposite to the displacement of the unit mass. So the work done by external agent to move unit mass from infinity to that point is always negative [APPLICATION]. So the gravitational potential at that point is also negative [CONCLUSION].

[CONCEPT: WD by force = force x displacement in the direction of the force]

(iii) Gravitational field strength, a *vector*, is given by the gradient of the potential graph. [CONCEPT]

Near the surface of the Earth, the net gravitational strength is directed *towards* the Earth. Near the surface of the Moon, the net gravitational strength is directed *towards* the Moon [APPLICATION].

Therefore, the gradient of the potential graph near the surface of the Earth and that near the surface of the Moon have *opposite* signs [CONCLUSION].

(iv) To reach the surface of the Earth, the spacecraft require a minimum kinetic energy that is just sufficient to overcome the gravitational potential energy difference between the surface of the moon and point P [APPLICATION].

Therefore, $\frac{1}{2}mv^2 = m(\phi_P - \phi_{Moon})$, where v is the minimum speed of the spacecraft.

[CONCEPT]

$$\therefore v = \sqrt{2(\phi_P - \phi_{Moon})} = \sqrt{2((-1.3) - (-3.9)) \times 10^6} = 2.28 \times 10^3 \text{ ms}^{-1}$$

- (d) (i)** To *escape* the planet means the mass must have sufficient kinetic energy at the surface of the planet to just reach infinity with zero speed at infinity [CONCEPT].

By Conservation of Energy,

Total Energy at the surface = Total Energy at infinity [CONCEPT]

$$\begin{aligned} \frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right) &= 0 + 0 \\ v &= \sqrt{\frac{2GM}{r}} \\ v &= \sqrt{\frac{2(6.67 \times 10^{-11})(1.20 \times 10^{24})}{\left(\frac{7.5 \times 10^6}{2}\right)}} \\ &= 6.53 \times 10^3 \text{ m s}^{-1} \end{aligned}$$

- (ii)** The average speed of the nitrogen gas is *higher* than the escape speed on this planet [APPLICATION], so most gases would have escaped and hence the planet does not have an atmosphere [CONCLUSION].

8(a)(i) Resistance $R = \rho l = \frac{(1.2 \times 10^{-5})(0.72)}{}$